

Zhao Zehui, Leila

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Mechanical Engineering undergraduate with experience in multidisciplinary design optimization, autonomous systems, and aerospace vehicle development. Interested in system engineering, space systems, robotics and UAVs.

EDUCATION

The University of Hong Kong, Bachelor's in Mechanical Engineering

2022 – now

- Course taken: Mechanics of Solid (A), Engineering Design and Manufacture (A+), Unmanned Aerial Vehicle design (A+), Dynamics and Control (A-), Thermodynamics (A-), Aircraft Stress Analysis and Finite Element (HKU MSc course) (A)
- Highest Attained Semester GPA: 3.36/4.3

WORK EXPERIENCES

The Hong Kong Productivity Council

Hong Kong

Winter Intern in Smart Manufacturing Division

Dec 2025 – Jan 2026

- Assisted in an ESG and textile recycling toolkit and developed survey by analyzing global regulations, supporting circular innovation for Hong Kong fashion SMEs and AI integration.
- Designed a 3D camera plus multi-axis robotic arm mounting system using SolidWorks (2D/3D drawings, BOM and GD&T).
- Developed applied AI for industrial thermographic anomaly inspection on 6-axes robotic arms with 86.4% precision.
- Designed a conceptual SLAM-based indoor drone inspection project proposal based on mechanical data (FEA) to detect internal building cracks, ranked 3rd in intern cohort pitching contest.

Faculty of Engineering, the University of Hong Kong

Hong Kong

Part-time Temporary Student Research Assistant

Sept 2025 – Apr 2026

- Mentored over 30 students in robotic arm control and design, FDM 3D printing, CAD software utilization, AI-assisted object identification and Raspberry Pi, enhancing their practical skills and understanding of engineering concepts.
- Provide constructive feedback to lecturers on behalf of workshop content which assisted them in workshop development.

Feelings Group Limited | Hong Kong Science & Technology Park

Hong Kong

Student Technician – Hardware team

Dec 2024 – Jan 2025

- Researched, analyzed and compared 3 critical radar systems and market database for medical devices tracking vital signs of small animals.
- Assisted in CCTV development for the medical device, using electronic modules, such as Raspberry Pi, LAN network and camera module.

The Hong Kong and China Gas Company Limited (Towngas)

Hong Kong

Summer intern in Residential Product & Quality department

Jul - Aug 2024

- Contributed to the development of 7 IoT gas appliances, and 10+ gateways.
- Identified potential problems in functional tests and provided effective suggestions to enhance user-friendliness of the mobile app interface.
- Assisted in the assessment of gas appliance accessories such as flexible hose and gas malleable iron fittings.
- Analyzed market potential for company's products and participated in daily meetings, contributed to strategic planning and understanding team structure and working principles and provided valuable feedback to supervisor regarding the market demand and daily workflow.

MAJOR PROJECTS & COMPETITIONS

HKU UAV Design and Drones Control

Sept 2026 - now

- Drone path planning algorithm study on MATLAB, including GFS, Dijkstra, GBFS, A* and RRT*.

TU Berlin Summer School 2026 - CANSAT Satellite Design, Build and Launch

Berlin, Germany

Software Lead & Embedded System Architect

Jul - Aug 2026

- Designed a four-stage CANSAT autonomous mission architecture integrating telemetry collection, parachute deployment logic, touchdown detection and post-landing navigation using Arduino.
- Rapidly prototyped an ESP32-CAM based colour tracking and conducted embedded vision testing for the navigation and performed feasibility studies under resource-constrained embedded hardware and validated the system through repeated real-world testing.
- Integrated sensors, microcontrollers and onboard electronics through soldering, hardware assembly and debugging, investigated onboard sensor-data logging and communication interfaces for autonomous mission development.

HKU Design, Build and Fly Team

Hong Kong, USA

HKU AIAA DBF 2026 Team Leader | Chief Engineer | Aerodynamic Lead | Observer Crew

Aug 2025 – May 2026

- Performed aerodynamic trade studies, including airfoil selection, CFD analysis, FEM and flight stability analysis using SolidWorks, XFLR5 and Ansys Fluent to optimize aircraft speed and performance, and validated simulation result with flight data from testing and past years' record, achieving 32 m/s.

- Designed, prototyped and tested 7 stow/deploy/tow/release mechanisms for a 4-meter-long banner weighed 500g, utilizing SolidWorks, FDM 3D printing, and Arduino.
- Led an engineering team and placed 32nd out of 98 global teams (3rd in Asia, 2nd in HK) in AIAA DBF 2026 competition.

HKU DBF Student Interest Group Leader

Sept 2024 – Jul 2026

- Secured HKD 100K funding from HKU Innovation Academy and multiple industry sponsorships.
- Managed a team of ~50 members, and allocated human resources for workshop training, manufacturing; Prepared administrative documents, promotion and train next year's leaders to enlarge the team size and development.

RoboMaster Tournament (Shenzhen)

Jul 2023

- Served as a mechanical team member and assisted in Robotic car assembly; Acquired knowledge in robotic arm design and working principles.

CERTIFICATES & AFFILIATIONS

HKU AIAA Student Chapter Founding Member & Student Advisor 2026 - 2027

Royal Aeronautical Society (HK) Student Member 2025 - 2026

IMechE Affiliated Member 2025 - 2026

HKIE Aircraft Division – Student Member 2025 – 2026

SKILLS & TECHNICAL EXPERTISE

Language: Chinese (Mandarin, Cantonese): Native, English: Proficient, German: Basic

CAD & Simulation: SolidWorks (proficient), AutoCAD (proficient), fusion360 (intermediate), Xflr5 (project experience), Ansys Fluent (project experience)

Programming: MATLAB (research experience), Arduino & ESP32 (project experience), Python (project experience)

Other Software: HIKIMICRO Studio (project experience), Bambu lab Studio (proficient), Microsoft Office Suite (proficient), Roboflow (project experience)